

HARITHA P H

M.Sc. Physics | Nanomaterials & Functional Materials

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EDUCATION

M.Sc. Physics

Mar Athanasius College (Autonomous) Kothamangalam
Mahatma Gandhi University, Kottayam Kerala, India

Jun 2024 – Aug 2026

B.Sc. Physics

Government Victoria College, Palakkad
University of Calicut, Kerala, India

Jun-2020 – Apr 2023

TECHNICAL SKILLS

Electrochemistry: Cyclic Voltammetry (CV), Galvanostatic Charge–Discharge (GCD), Electrochemical Impedance Spectroscopy (EIS), Coin-cell Fabrication

Materials Synthesis & Electrode Fabrication: Microwave-assisted Exfoliation, Electrophoretic Deposition, Electrochemical Deposition, Sol–Gel, Co-precipitation

Materials Characterization: X-ray Diffraction (XRD), Raman Spectroscopy, Scanning Electron Microscopy (SEM), FTIR, UV–Visible Spectroscopy

Software: OriginPro, Python, LaTeX, MS Office, EC-Lab (BioLogic)

RESEARCH INTERESTS

- Electrochemical Materials & Energy Storage
- Electrode–Electrolyte Interfaces
- Two-Dimensional Materials
- Functional Nanomaterials
- Thin Films & Surface Engineering
- Advanced Materials Characterization

RESEARCH EXPERIENCE

Master's Research Project

IISER Thiruvananthapuram | Shaijumon Research Group

Mar 2026 – Jul 2026

Electrochemical Synthesis of Transition Metal Dichalcogenide-Based Materials for Supercapacitor Applications

- Synthesized mixed-phase MoS_2 nanosheets using microwave-assisted exfoliation.
- Optimized electrophoretic deposition to fabricate binder-free thin-film electrodes.
- Characterized materials using XRD, Raman spectroscopy and SEM.
- Performed electrochemical characterization using CV, GCD and EIS.
- Investigated the influence of material and electrode preparation on electrochemical performance.
- Explored direct electrochemical deposition of TMD-based materials.

Bachelor's Research Project

Govt Victoria College | NiFe_2O_4 Ferrite Nanoparticles

Feb 2023 – Apr 2023

Synthesis and Characterization of NiFe_2O_4 Ferrite Nanoparticles

- Synthesized nanoparticles using sol–gel and co-precipitation methods.
- Characterized materials using XRD, FTIR and UV–Visible spectroscopy.

INTERNSHIPS

Science Communication Intern

May – Jun 2025

Centre for Science in Society (CUSAT)

- Participated in the International Year of Quantum Science and Technology (IYQ 2025).
- Developed science communication and outreach skills.

Solar Energy Intern

7–15 May 2025

Agency for New and Renewable Energy Research and Technology (ANERT)

- Gained exposure to photovoltaic technologies and recent developments in solar panels.

ACADEMIC & OTHER ACHIEVEMENTS

- Graduated first in the class (Gold medalist), Mar Athanasius College (Autonomous), Mahatma Gandhi University (2026).
- NCC Senior Under Officer and Republic Day Camp (RDC) Participant, a national-level NCC camp emphasizing leadership, discipline and teamwork.

REFERENCES

Prof. M. M. Shaijumon

Professor and Head, School of Physics
Indian Institute of Science Education and Research
(IISER) Thiruvananthapuram Kerala, India
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Academic Referee & Internal Project Supervisor

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Academic Referee